CSE273:
Foundations of Machine Learning

UNIT-4 Part-3
(Eigenvalues & Eigenvectors)

Course Instructor
Sumit Mittu

Eigen Values:

- **Definition:** Scalars linked to linear systems.
- **Usage:** Used in solving matrix equations.
- **Origin:** 'Eigen' means 'characteristic' in German.
- **Alternate Names:** Characteristic roots or latent roots.
- **Role:** Scale eigenvectors during transformations.

It is represented as:

$$Av = \lambda v$$

Where,

- *A is the matrix,*
- *v is associated eigenvector, and*
- *λ is scalar eigenvalue.*

Procedure for finding Eigenvalues:

- Let A be n*n matrix.
- Find eigenvalues λ of A by:

$$det(A - \lambda I)$$

Key points about eigen values:

- **Eigenvalue:** Factor determining stretching magnitude.
- **Negative Eigenvalue:** Reverses the transformation's direction.
- **Real Matrices:** Always have real eigenvalues.
- **Complex Matrices:** Eigenvalues may sometimes be complex.
- **Significance:** Linked to the fundamental algebra theorem.

Problem 1: Compute eigen value for A.

$$\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

det(A−λI)=0

Where:

- λ= eigenvalue (scalar)
- I= identity matrix

Subtract λI

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 2 \\ 5 & 4-\lambda \end{bmatrix}$$

$$\begin{vmatrix} 1-\lambda & 2 \\ 5 & 4-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(4-\lambda) - (2 \times 5) = 0$$

$$(1-\lambda)(4-\lambda) = 4 - \lambda - 4\lambda + \lambda^2$$

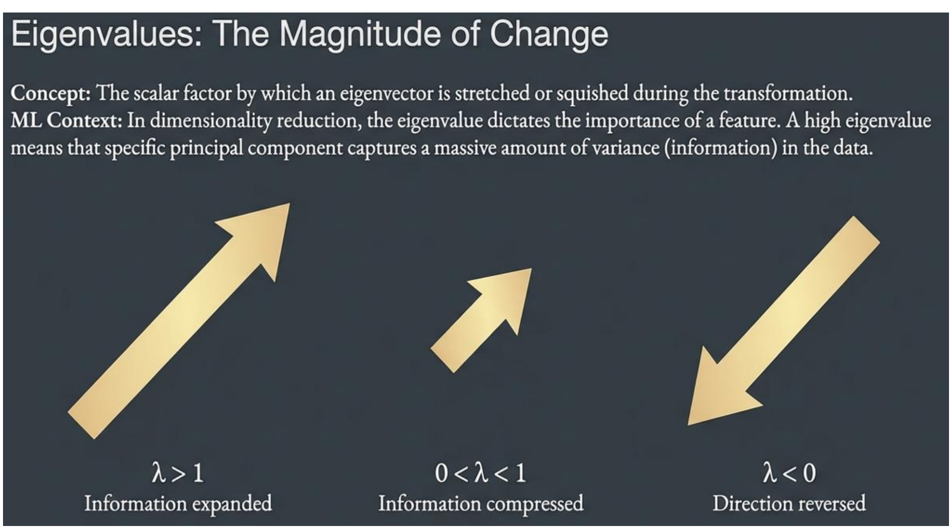
So:

$$4 - 5\lambda + \lambda^2 - 10 = 0$$

$\lambda^2 - 5\lambda - 6 = 0$

$$(\lambda - 6)(\lambda + 1) = 0$$

$\lambda = 6, \lambda = -1$



Problem 2:

Let $A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 4 & 7 \\ 0 & 0 & 6 \end{bmatrix}$. Find the eigenvalues of A.

Eigen Vectors:

- Vector unaffected by linear transformation.
- Scaled version of original vector.
- Associated with specific eigenvalue.
- Indicates direction of transformation.

Procedure for finding eigenvalues and eigenvectors:

- Let A be n*n matrix.
- Find eigenvalues λ of A by:

$$det(A - \lambda I)$$

- For each λ, find eigenvectors by:

$$(A\text{-}\lambda I)v = 0$$

- To verify your work, make sure that AX= λx for each λ.

Problem 1:

- $(A - \lambda I)\ v = 0$
- Where:
- $v = \begin{bmatrix} a \\ b \end{bmatrix}$)eigenvector)
- **Substitute** $\lambda = 6$
- $A - 6I = \begin{bmatrix} 1-6 & 2 \\ 5 & 4-6 \end{bmatrix} = \begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix}$
- $\begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

From matrix multiplication:

- $-5a + 2b = 0$
- $5a - 2b = 0$
- Both equations are **same (dependent)**

5a=2b

$$b = \frac{5}{2}a$$

Let:

$$a = 2$$

Then:

$$b = 5$$

Final Eigenvector

$$v = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

For λ = -1

$$\Rightarrow \begin{bmatrix} 1-(-1) & 2 \\ 5 & 4-(-1) \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} 0 =$$

$$\Rightarrow \begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} 0 =$$

$\Rightarrow 2a + 2b = 0, \Rightarrow 5a + 5b = 0$

simplifying the above equation we get, $a = -b$

The required eigenvector is, $\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Problem 2: Find the eigen values and vector for:

$$A = \begin{bmatrix} -5 & 2 \\ -7 & 4 \end{bmatrix}$$

Problem 3: Find the eigen values and vector for:

$$\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

Problem 4: Find the eigen values and vector for:

$$A = \begin{bmatrix} 5 & -10 & -5 \\ 2 & 14 & 2 \\ -4 & -8 & 6 \end{bmatrix}$$

```
import numpy as np

A = np.array([[1, 2],
              [5, 4]])

eigenvalues, eigenvectors = np.linalg.eig(A)
```

`np.linalg.eig()` returns:

Eigenvalues (1D array)

Eigenvectors (columns of matrix)

```
from scipy.linalg import eig

eigenvalues, eigenvectors = eig(A)
```

→ **For symmetric matrices, use:**
np.linalg.eig(A)

Applications of Eigenvalues and Eigenvectors:

- Eigenvalues and eigenvectors are used in Principal Component Analysis for reducing dimensionality while preserving important information
- They help in feature extraction by transforming correlated features into uncorrelated ones
- Useful in noise reduction by ignoring components with small eigenvalues
- Enable data visualization by projecting high-dimensional data into 2D or 3D space

Summary:

Concept	Dimension	NumPy Syntax	Primary ML Role
Scalar	0D	`np.array(x)`	Hyperparameters, Loss
Vector	1D	`np.array([x, y])`	Feature sets, Weights
Matrix	2D	`np.array([[...]])`	Datasets, Neural Layers
Tensor	ND	`np.random.rand(...)`	Image/Video Batches
Eigen	Transform	`np.linalg.eig(M)`	PCA, Dimensionality Reduction